

Mingxiao Li

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RESEARCH EXPERIENCE

KU Leuven

Postdoctoral Researcher in Computer Science

Leuven, Belgium

Nov. 2024 - present

- Advisor: Prof. Marie-Francine Moens
- Research Interests: Vision-Language, Generative Models, Large Vision-Language Model

EDUCATION

KU Leuven

Ph.D. in Computer Science

Leuven, Belgium

Sept. 2019 - Nov. 2024

- Advisor: Prof. Marie-Francine Moens
- Thesis: Leveraging External Knowledge in Deep Learning and Conditional Generative Models.
- **Awarded summa cum laude with the congratulations of the examination committee (greatest distinction), ranked in the top 5% of the Faculty of Engineering Science.**

KU Leuven

Master of Artificial Intelligence

Leuven, Belgium

Sept. 2018 - Sept. 2019

KU Leuven

Master of Quantum Chemistry and Computational Modeling

Leuven, Belgium

Sept. 2016 - Jun. 2018

East China University of Science and Technology

Bachelor of Material Physics

Shanghai, China

Sept. 2011 - Jun. 2015

- *Academic scholarship:* Academic year 2011-2012, 2012-2013, 2013-2014, 2014-2015
- *Honors Graduate:* (Top 5% students in Department of Material)

RESEARCH INTEREST

Vision-Language Models are critical for constructing AI systems that can communicate with humans. My previous research focused on knowledge-based visual question answering and visual dialogue. Additionally, I have explored the use of vision-language models in embodied AI, where we leverage text-to-image models to assist agents in decision-making. (This work was recognized as an oral presentation at AAAI)

Visual Generative Models are crucial components in the development of AGI. My prior research focuses on advancing the sampling algorithm of diffusion model through shifting time steps, and on designing controllable video generation with diffusion model, which involved animating images in accordance with customized object motions to align with user expectations.

LLM & MLLM are crucial for developing general AI systems. Beyond image processing, enabling LLMs to understand videos is also important. I recently initiated a project aimed at enhancing LLMs with the capability to comprehend extended videos.

SELECTED PUBLICATIONS

- **Mingxiao Li***, Bo Wan*, Marie-Francine Moens & Tinne Tuytelaars. Animate Your Motion: Turning Still Images into Dynamic Videos. *European Conference on Computer Vision (ECCV 2024)*. [Paper] [Code]
- **Mingxiao Li***, Tingyu Qu*, Ruicong Yao, Wei Sun, & Marie-Francine Moens. Alleviating Exposure Bias in Diffusion Models through Sampling with Shifted Time Steps. *International Conference on Learning Representations (ICLR 2024)*. [Paper] [Code (DDPM ver.)][Code (ADM ver.)]
- Mang Ning, **Mingxiao Li ***, Jianlin Su *, Haozhe Jia, Lanmiao Liu, Martin Beneš, Albert Ali Salah, Itir Onal Ertugrul. DCTdiff: Intriguing Properties of Image Generative Modeling in the DCT Space. *International Conference on Machine Learning (ICML 2025)*. [Paper] [Code]

- Mang Ning, **Mingxiao Li**, Jianlin Su, Albert Ali Salah & Itir Onal Ertugrul. Elucidating the Exposure Bias in Diffusion Models. *International Conference on Learning Representations (ICLR 2024)*. [Paper] [Code]
- Jingyuan Sun*, **Mingxiao Li***, Zijiao Chen, Yunhan Zhang, Shaonan Wang, & Marie-Francine Moens. Contrast, Attend and Diffuse to Decode High-Resolution Images from Brain Activities. *Advances on Neural Information Processing Systems. (NeurIPS 2023)*. [Paper] [Code]
- **Mingxiao Li***, Zehao Wang*, Tinne Tuytelaars, & Marie-Francine Moens. Layout-Aware Dreamer for Embodied Visual Referring Expression Grounding. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2023)* (**Oral Presentation**). [Paper] [Code]
- **Mingxiao Li**, Marie-Francine Moens. Dynamic Key-value Memory Enhanced Multi-step Graph Reasoning for Knowledge-based Visual Question Answering. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2022)*. [Paper][Code]
- Jingyuan Sun*, **Mingxiao Li***, Zijiao Chen, Marie-Francine Moens. NeuroCine: Decoding Vivid Video Sequences from Human Brain Activities. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2025)* (**Oral Presentation**).[Paper]

SELECTED PREPRINT

- **Mingxiao Li**, Tingyu Qu, Tinne Tuytelaars, Marie-Francine Moens. Towards More Accurate Personalized Image Generation: Addressing Overfitting and Evaluation Bias [Code]
- Tingyu Qu, **Mingxiao Li**, Tinne Tuytelaars, Marie-Francine Moens. TS-LLaVA: Constructing Visual Tokens through Thumbnail-and-Sampling for Training-Free Video Large Language Models [Paper][Code]
- Maria Mihaela Trusca, **Mingxiao Li**, Marie-Francine Moens. Action-based image editing guided by human instructions. [Paper]

* denotes equal contribution

AWARD

KU Leuven PhD graduate with summa cum laude (highest distinction), ranked in the top 5% of the Faculty of Engineering Science (2024)

European Union Erasmus Mundus Full Scholarship. (2016-2018)

Chinese Government Scholarship for Outstanding Undergraduate Students. (2015)

SERVICES

Reviewer: ICML, CVPR, ICLR, NeurIPS, EMNLP, ACL, AAAI, EACL, ECML, ECAI

Area Chair: ACL Rolling Review (2025.02)

Workshop Organizer: AAAI-2024 Workshop: Artificial Intelligence for Brain Encoding and Decoding (AIBED)

LIST OF PUBLICATIONS

(Sort in chronological order)

- Mang Ning, **Mingxiao Li** *, Jianlin Su *, Haozhe Jia, Lanmiao Liu, Martin Beneš, Albert Ali Salah, Itir Onal Ertugrul. DCTdiff: Intriguing Properties of Image Generative Modeling in the DCT Space. *International Conference on Machine Learning (ICML 2025)*. [Paper] [Code]
- Zehao Wang*, **Mingxiao Li***, Minye Wu, Marie-Francine Moens & Tinne Tuytelaars. nstruction-guided path planning with 3D semantic maps for vision-language navigation. *Neurocomputing* [Paper] [Code]

- Jingyuan Sun*, **Mingxiao Li***, Zijiao Chen, Marie-Francine Moens. NeuroCine: Decoding Vivid Video Sequences from Human Brain Activities. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2025)* (**Oral Presentation**). [Paper]
- **Mingxiao Li***, Bo Wan*, Marie-Francine Moens & Tinne Tuytelaars. Animate Your Motion: Turning Still Images into Dynamic Videos. *European Conference on Computer Vision (ECCV 2024)*. [Paper] [Code]
- Sun, Wei, **Mingxiao Li***, Jingyuan Sun, & Marie-Francine Moens. DMON: A Simple yet Effective Approach for Argument Structure Learning. *Proceedings of LREC-COLING 2024 - The 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation, Page 5109–5118* [Paper]
- Jingyuan Sun, Mingxiao Li, Cartuyvels Ruben & Marie-Francine Moens. ERC Advanced Grant Project CAL-CULUS: Extending the Boundary of Machine Translation. *Proceedings of Association for Machine Translation, Page 16-17* [Paper]
- **Mingxiao Li***, Tingyu Qu*, Ruicong Yao, Wei Sun, & Marie-Francine Moens. Alleviating Exposure Bias in Diffusion Models through Sampling with Shifted Time Steps. *International Conference on Learning Representations (ICLR 2024)*. [Paper] [Code (DDPM ver.)][Code (ADM ver.)]
- Mang Ning, **Mingxiao Li**, Jianlin Su, Albert Ali Salah & Itir Onal Ertugrul. Elucidating the Exposure Bias in Diffusion Models. *International Conference on Learning Representations (ICLR 2024)*. [Paper] [Code]
- Jingyuan Sun*, **Mingxiao Li***, Zijiao Chen, Yunhan Zhang, Shaonan Wang, & Marie-Francine Moens. Contrast, Attend and Diffuse to Decode High-Resolution Images from Brain Activities. *Advances on Neural Information Processing Systems. (NeurIPS 2023)*. [Paper] [Code]
- Jingyuan Sun*, **Mingxiao Li***, Zijiao Chen, & Marie-Francine Moens. Decoding Realistic Images from Brain Activity with Contrastive Self-supervision and Latent Diffusion. *Proceedings of the 26th European Conference on Artificial Intelligence* [Paper][Code]
- **Mingxiao Li***, Zehao Wang*, Tinne Tuytelaars, & Marie-Francine Moens. Layout-Aware Dreamer for Embodied Visual Referring Expression Grounding. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2023)* (**Oral Presentation**). [Paper] [Code]
- **Mingxiao Li**, Marie-Francine Moens. Dynamic Key-value Memory Enhanced Multi-step Graph Reasoning for Knowledge-based Visual Question Answering. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2022)*. [Paper][Code]
- **Mingxiao Li**, Marie-Francine Moens. Modeling Coreference Relations in Visual Dialog. *The 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2021)*. [Paper][Code]
- Jaron Maene, **Mingxiao Li**, & Marie-Francine Moens. Towards understanding iterative magnitude pruning: Why lottery tickets win. *Sparsity in Neural Networks 2021 Workshop* [Paper]

Preprint

- **Mingxiao Li**, Tingyu Qu, Tinne Tuytelaars, Marie-Francine Moens. Towards More Accurate Personalized Image Generation: Addressing Overfitting and Evaluation Bias [Code]
- Tingyu Qu, **Mingxiao Li**, Tinne Tuytelaars, Marie-Francine Moens. TS-LLaVA: Constructing Visual Tokens through Thumbnail-and-Sampling for Training-Free Video Large Language Models. [Paper][Code]
- Maria Mihaela Trusca, **Mingxiao Li**, Marie-Francine Moens. Action-based image editing guided by human instructions. [Paper]